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AICOR Institute for Artificial Intelligence
WiSe 2026/2027
Worksheet 3

Exercises for the Lecture Advanced Methods of AI

Asking, Defining, Constraining, Deriving, Generating

Tutorial: Thursday, 12.11.2026, 8:30–10:00, room 0.30, ECO 5

Last week you *represented* the pizza world in first-order logic and in Python. Tonight a service robot in a pizzeria *uses* that knowledge. Everything it does starts with one kind of expression, a **description**, written in the Entity Query Language (EQL). The five acts of the evening are five things the robot does with one:

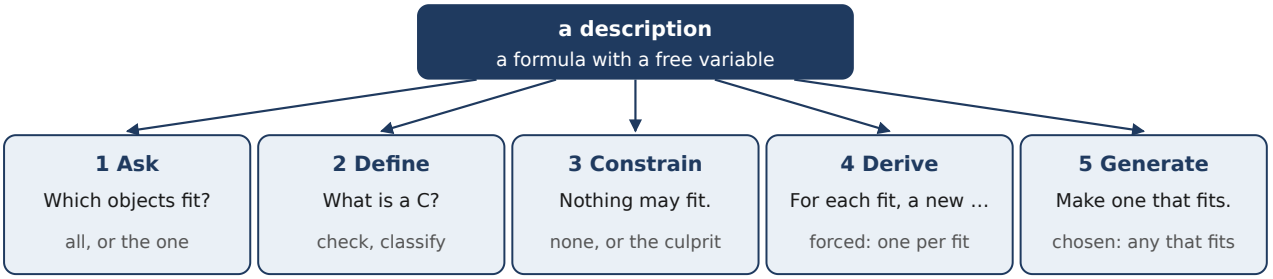


Figure 1. One description, five uses. The expression stays the same; what changes is what is done with it. Asking and constraining *read* what the robot knows; defining, deriving and generating *write* to it.

How to work

The evening happens in the notebook `03-pizza_queries.ipynb` (link on Stud.IP). Open it in pairs, run its setup cell, and work through it from top to bottom.

Three times, the tutor calls a **pen break**. Put the laptop aside and turn to this sheet: write the logic by hand for about six minutes. Then the tutor writes an answer on the board and takes your questions, and you go back to the notebook.

PEN BREAK 1 · AFTER ACT 1

Asking

About 6 minutes of writing, then the board.

(a) Every topping of a pizza without toppings

```
Q1 = a(Pizza)(toppings=every_is(VegetableTopping)).from_(pizzas)
```

Write Q1 as a formula with the free variable *x*. The robot returned plain_1, a pizza without any topping: show with your formula why plain_1 satisfies it.

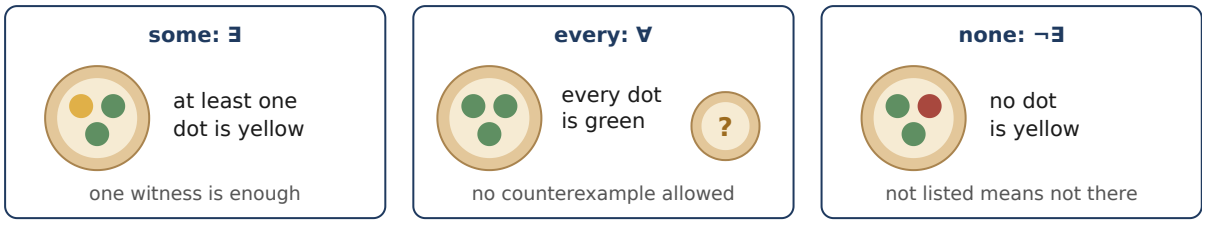


Figure 2. The three quantifier patterns, as the robot checks them on one pizza. The small pizza has no toppings at all: is every dot on it green?

(b) Making a description precise

“The pizza with olives” fitted two pizzas; Ben wants the one *without fish*. Write the precise description as a formula with the free variable *x*.

PEN BREAK 2 · AFTER ACT 3

Defining and constraining

About 6 minutes of writing, then the board.

(a) Concluding from a name

The menu's definition from the notebook:

$$\forall x\ (\text{VegetarianPizza}(x) \leftrightarrow \text{Pizza}(x) \land \neg\exists t\ (\text{hasTopping}(x, t) \land (\text{MeatTopping}(t) \lor \text{FishTopping}(t))))$$

A pizza from the other branch arrives with a green V; nobody wrote down its toppings. What may the robot tell Chidi, who avoids fish? And Ben, who avoids milk? Which direction of $\leftrightarrow$ do you use, and which direction did the robot use in Act 2 to put the green V on tonight's pizzas?

(b) A constraint is a query that must have no answer

The kitchen's rule: every pizza of an order is safe for the guest of that order. Write it as a formula (with hasPizza, guest and IsSafeFor), and then its negation, with the ¬ pushed inwards. Which of the two did `dangerous` ask?

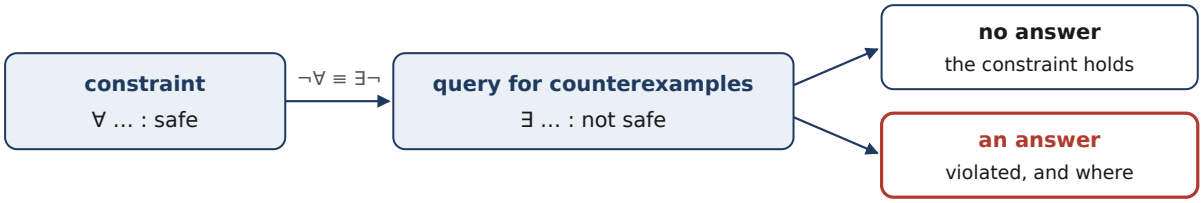


Figure 3. A universal statement is violated exactly when a counterexample exists.

PEN BREAK 3 · AFTER ACT 5

Deriving, and the whole evening

About 4 minutes for (a); the summary together with the tutor.

(a) A rule with ∃ in its head

The milk rule of Act 4 is

```
∀x (Pizza(x) ∧ ∃t (hasTopping(x, t) ∧ CheeseTopping(t)) → ∃w (Warning(w) ∧ about(w, x) ∧ allergen(w) = milk))
```

Underline its body and its head. The label rule of Act 2, $\forall x\,(\varphi(x) \rightarrow \text{VegetarianPizza}(x))$, only labelled pizzas that were there; the milk rule created five new warnings. What in the formula makes the difference?

(b) The whole evening

For each use of a description, say what it is in logic.

Use	Tonight's example	In logic
1 Ask	the cheesy pizzas; *the* pizza with olives	
2 Define	the green V	
3 Constrain	no pizza unsafe for its guest	
4 Derive	the allergen warnings	
5 Generate	Dana's pizza	

Take home

One expression, a description, carried the whole evening. Its meaning is a formula with free variables; what you *do* with it decides whether it is a question, a definition, a constraint, the body of a rule or a specification, and which backend answers it decides whether it is checked, computed or constructed.

To think about at home: a robot tidies a kitchen. Give one description for each of the five uses, and the backend that should answer it.